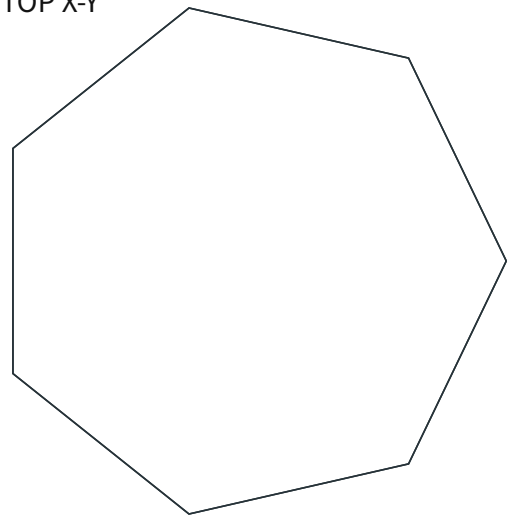


MFG-SP-SH-01 • v0.8 MANUFACTURING DRAWING

TOP X-Y



FRONT X-Z



SIDE Y-Z



Drawing MFG-SP-SH-01 | Part SP-SH-01 — Spool spindle | Qty 1

Rev final-design-fabrication-closure-v0.8 | Units mm | Scale NTS, dimensions govern | Third-angle projection | source commit df59fd138780f538a5fa62dad6e2770b49bf4d94

Material: Ø12 h6 S45C | Process/finish: cut/turn faces + collar flats

Overall: X 12.000 × Y 12.000 × Z 143.000 mm | general tolerance ISO 2768-m unless noted

Critical: Ø12 h6 x143; straightness0.05; two 6001 bearings; axial collars carry spool load

GD&T/datums: A=common journal axis; B=drive-end shoulder/face; C=keyway or retaining feature plane; verify datum marking on supplier inspection report

Inspection: CMM/caliper against STEP and drawing; deburr and visual crack/warp inspection; micrometer bearing seats; dial-indicator TIR/runout between centres

Edges: remove burrs; break unspecified sharp edges C0.3–0.5. No unapproved material/tolerance substitution.

Release: PASS — physical validation NOT_RUN; procurement and machining USER_APPROVAL_REQUIRED.